Fraud Team 4

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```
library(lubridate)
```

```
## Warning: package 'lubridate' was built under R version 3.5.3
```

```
##
## Attaching package: 'lubridate'
```

```
## The following object is masked from 'package:base':
##
##     date
```

```
library(xts)
```

```
## Warning: package 'xts' was built under R version 3.5.3
```

```
## Loading required package: zoo
```

```
## Warning: package 'zoo' was built under R version 3.5.3
```

```
##
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```
library(dygraphs)
```

```
## Warning: package 'dygraphs' was built under R version 3.5.3
```

```
library(readxl)
```

```
## Warning: package 'readxl' was built under R version 3.5.3
```

#Reading the data

```
Prices=read_xls("C:/Users/talga/Desktop/Masters/R/MyDataFiles/HW_sat.xls",
                 sheet = "Sheet2", col_names=FALSE)
```

```
## New names:
## * `` -> ...1
```

```
length(Prices)
```

```
## [1] 1
```

```
head(Prices)
```

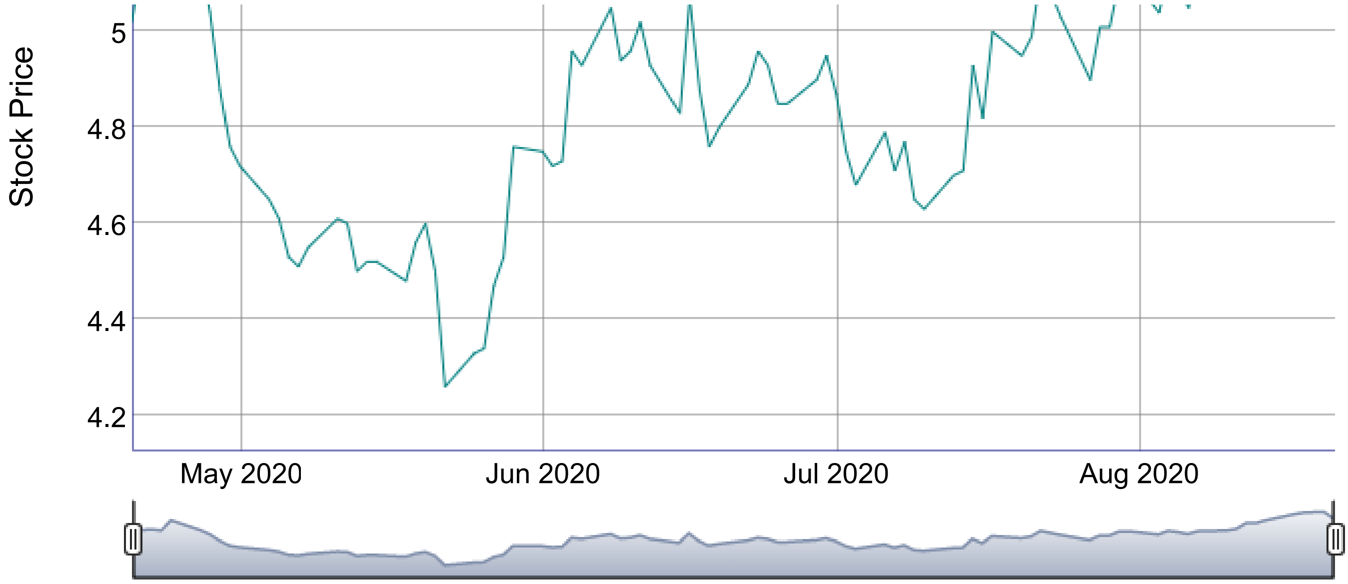
```
## # A tibble: 6 x 1
##    ...1
##   <dbl>
## 1  5.02
## 2  5.15
## 3  5.17
## 4  5.14
## 5  5.4
## 6  5.15
```

#Creating some time series with 90 days, except saturdays and sundays:

```
Imaginary_Dates=seq.Date(from = as.Date("2020-04-20"),
                         to = as.Date("2021-04-20"),
                         by = 1)

Imaginary_Dates=Imaginary_Dates[-seq(from=7, to=length(Imaginary_Dates), by=7)]

Imaginary_Dates=Imaginary_Dates[-seq(from=6, to=length(Imaginary_Dates), by=6)]

Imaginary_Dates=Imaginary_Dates[1:90]
```

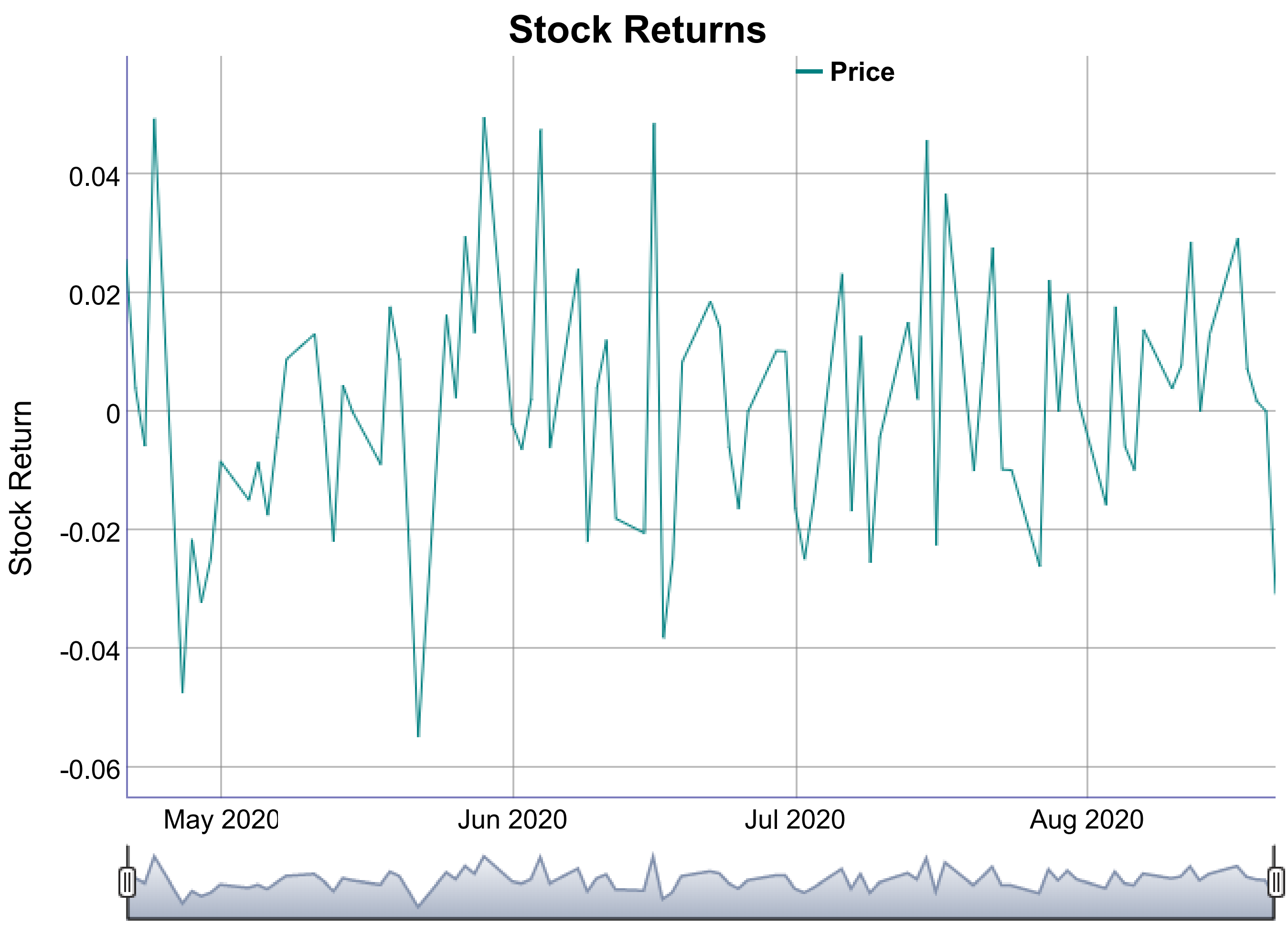
#Merging dates and Prices

```
Prices=xts(Prices, order.by = Imaginary_Dates); colnames(Prices)="Price"

dygraph(Prices, main = "Stock Prices") %>%
  dyRangeSelector() %>%
  dyLegend(show = "always", hideOnMouseOut = TRUE) %>%
  dyAxis("y", label = "Stock Price") %>%
  dyHighlight(highlightCircleSize = 5,
              highlightSeriesOpts = list(strokeWidth = 4)) %>%
  dyOptions(axisLineColor = "navy", gridLineColor = "grey")
```



#Checking the fraud possibility

```
Returns=na.omit(diff(log(Prices)))

dygraph(Returns, main = "Stock Returns") %>%
  dyRangeSelector() %>%
  dyLegend(show = "always", hideOnMouseOut = TRUE) %>%
  dyAxis("y", label = "Stock Return") %>%
  dyHighlight(highlightCircleSize = 5,
              highlightSeriesOpts = list(strokeWidth = 4)) %>%
  dyOptions(axisLineColor = "navy", gridLineColor = "grey")
```



```
Tuesday=ifelse(.indexwday(Returns)==2,
                       1,
                       0)

Fraud_check=lm(Returns$Price~Tuesday)

summary(Fraud_check)
```

```
##
## Call:
## lm(formula = Returns$Price ~ Tuesday)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.054449 -0.014583  0.000359  0.013017  0.049885
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.0003588  0.0025241  -0.142    0.887
## Tuesday      0.0062380  0.0056126   1.111    0.269
##
## Residual standard error: 0.02127 on 87 degrees of freedom
## Multiple R-squared:  0.014,  Adjusted R-squared:  0.002666
## F-statistic: 1.235 on 1 and 87 DF,  p-value: 0.2694
```

We can conclude that there is insignificant effect of Tuesday trading

on the stock returns. Hence, there is no evidence of fraud.